

RoadWatch.io

Intelligent Road Monitoring & Governance

Presented by The Outlier - Musa Qureshi

The Problem: Broken Roads, Broken Systems

Unsafe Infrastructure

Deteriorating roads cause accidents, slow traffic, and cost cities billions annually in emergency repairs.

Opaque Reporting

Manual complaint systems are slow and invisible — citizens have no way to track if their reports are acted upon.

Data Blindness

Municipalities lack real-time visibility into road conditions, making proactive maintenance nearly impossible.

1.19M+

Accident Tracking

Road defects increase crash risk, especially in high-traffic urban corridors.

70%+

Repair Delays

Manual reporting can leave potholes unresolved for days or weeks.

30-40%

Budget Waste

Reactive fixes are significantly more expensive than preventive maintenance.

Without timely reporting and visibility, cities spend more, respond slower, and leave drivers exposed to avoidable risk.

Our Solution: End-to-End Road Intelligence



RoadWatch AI uses **computer vision** to detect road damage from user-submitted images, automatically scores severity, and routes each issue to the right authority — with an emergency layer for critical hazards.

→ Detect

AI identifies and segments potholes, cracks, and surface damage in real time.

→ Prioritize

Severity scoring ensures the most dangerous defects are addressed first.

→ Coordinate

Automated workflows assign the right agency and track repairs to resolution.

Top 10 Features



AI Severity Scoring

Auto-classifies damage severity and repair priority.



Assignment Workflow

Tracks each issue from report through resolution.



Repair Cost Estimation

Calculates expected maintenance cost per issue.



RoadSoS Emergency Layer

Triggers alerts to emergency services for critical hazards.



Road Risk Heatmap

Visual map of accident-prone or damaged areas.



Authority Copilot

Suggests responsible agency and repair plan.



Predictive Failure

Forecasts future road wear and risks.



Road Health Index

Dashboard metric for overall road conditions.



Public Transparency Dashboard

Shows citizens live status of reported issues.



Model Benchmarking

Validates and reports on AI model performance.

Services & Modules

RoadWatch AI's backend is organized into seven specialized services, each handling a distinct layer of the platform — from image capture to emergency response.

1 **Reporting Service**
Captures images, geotags, and logs new road issues.

2 **Detection Service**
Runs YOLO-Seg AI models to identify potholes and damage.

3 **Governance Service**
Manages authority assignment, status updates, and communication.

4 **Prediction Service**
Estimates future failure risk and computes repair costs.

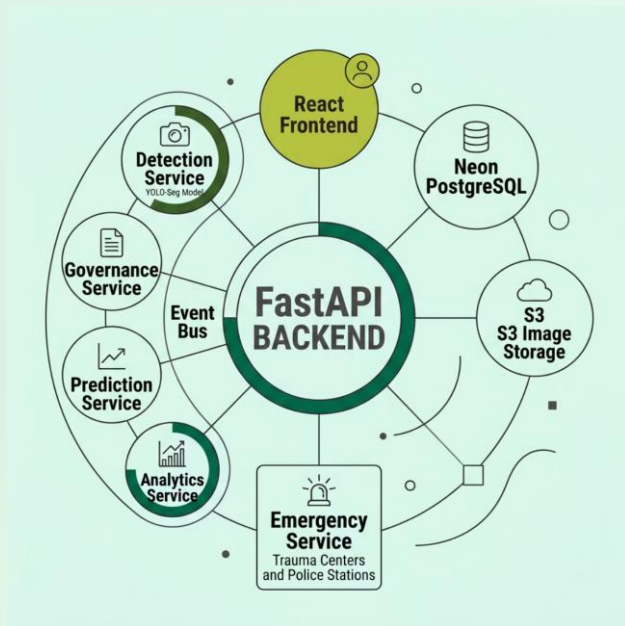
5 **Emergency Service**
Implements RoadSoS logic to notify hospitals and police.

6 **Analytics Service**
Generates dashboards, metrics, and heatmaps.

7 **Notification Service**
Sends alerts to citizens and officials via email and SMS.

System Architecture

RoadWatch AI uses a **modular monolithic architecture** with an internal event-driven workflow. The React frontend communicates with a FastAPI backend, which publishes tasks to an internal Event Bus — triggering each service independently without the complexity of an external message queue.



React + TypeScript

Responsive citizen and authority frontend

FastAPI Backend

High-performance Python API layer

Neon PostgreSQL

Cloud-native, scalable data storage

S3 Image Storage

Secure, durable media repository

Model Selection: Why YOLO-Seg

Selection Criteria

We benchmarked leading models on speed, detection accuracy, segmentation quality, and ease of integration.

- ✔ **YOLO-Seg won** because it delivers real-time pixel-level segmentation (~30 FPS) — essential for estimating repair area — while maintaining competitive accuracy. Heavy two-stage models were excluded for lacking real-time inference.

Model	Speed	mAP	IoU	Verdict
YOLOv8	~50 FPS	~55%	—	Strong detector
YOLO-Seg ✓	~30 FPS	~50%	~60%	Selected
Efficient Net	~5 FPS	~40%	—	Not chosen
Faster R-CNN	~5 FPS	~60%	—	Too slow
RT-DETR	~15 FPS	~50%	—	Not finalized

Evaluation Metrics

Results from YOLO-Seg testing on urban road imagery demonstrate strong real-world performance across detection and segmentation benchmarks.

0.85

Precision

Low false-positive rate on damage detection

0.80

Recall

High capture rate on severe defects

0.825

F1 Score

Balanced precision-recall performance

0.60

IoU

Mask overlap quality for area estimation

0.75

Dice Score

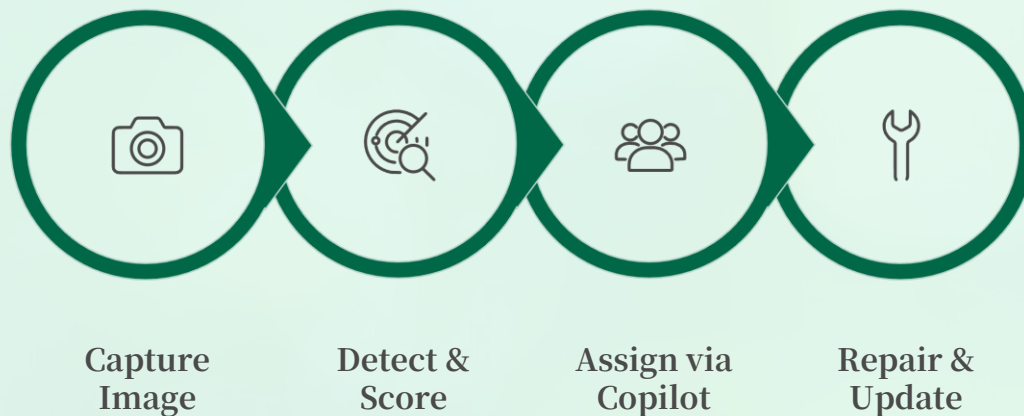
Segmentation accuracy on road damage

50ms

Latency

Per-image inference time for responsiveness

Methodology & Workflow



From capture to resolution, every step is automated and tracked — with an emergency bypass for life-threatening defects.

Data Pipeline

Urban road images collected, annotated for potholes and cracks, and augmented for robust training.

Training & Deployment

YOLO-Seg fine-tuned via cross-validation for best F1. FastAPI serves the model; React displays results; periodic analytics jobs refresh dashboards.

Impact, Viability & Next Steps

Safer Roads

Faster fixes reduce accidents. Citizens gain trust through live transparency dashboards.

Smarter Planning

Data-driven insights improve city maintenance budgets and long-term infrastructure strategy.

Built to Scale

Cloud-native stack (React, FastAPI, NeonDB) scales to any city. Open-source foundation keeps costs low.

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Demo Video

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Thank You

For your time and attention to RoadWatch.io.

Driving Safer Roads, Smarter Cities.

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